Elixir cheat sheet

drum machine firmware for the M-VAVE FM-1 · 1 of 2

Black keys numbered 1–11 and white keys 1–16, left to right

A BEAT IN A MINUTE

HOME grid · black **1** (D1) · white 1 5 9 13 · black **4** (SN) · white 5 13 · black **6** (CH) · white 1 3 5 … 15 · **PLAY**. Turn **PRESETS** for kits, **OCT+** loads one.

KEYS AND KNOBS

White 1–16	Steps of the bar on screen · patterns (pattern page) · rolls and effects (PERF)
Black 1–7	Track: D1 D2 D3 SN CY CH OH (CH and OH share a voice)
Black 8–11	Bar 1–4 · banks on pattern, song, save and load pages · repeat length in PERF
OCT–	**SHIFT** (hold): the screen shows what SHIFT (OCT–) + each key does
OCT+	**COPY** (hold) · OK in dialogs · casts a spell
ALGORITHM	Select the track
SELECT	The page; in lists, the row
PRESETS	Kit browser; on a sound page, the sound browser
KNOB 1–4	The four cards at the bottom of the screen
MASTER	Output volume

BUTTONS (PRESS AGAIN FOR THE NEXT PAGE)

HOME	Grid → patterns → song
EDIT	Model → oscillator → FM → transient
ENV	Amp EG → gate → mod EG / velocity
SCL	Filter → shape → insert FX
LFO	LFO 1 → LFO 2 → random per hit → cross mod / choke
GLO	Mixer → config
FX	FX bus → reverb (SELECT for the pages)
SEQ	Track pages (track, generator, action, groove, bounce, melody, scale) → step edit
ARP	PERF: PERFORM → SPELLS → GLOBAL MOD
SAVE	Save → load
PLAY · REC	Start / stop · record on / off

SHIFT (OCT–) + …

white 1–7	Play that track's voice
white 8 · 9	KIT: load a kit · MORPH: load the morph target
white 10 · 16	SONG page · PATTERN page
white 11 · 12	PROJECT: load a project · CONFIG
white 13	RELOAD the kit and pattern as saved
white 14	LAST: tap the track's last step (again: all tracks)
white 15	FX
track key	Mute / unmute
bar key	Track length 16 / 32 / 48 / 64
OCT+	**CLEAR**: SELECT a choice, SHIFT (OCT–) + OCT+ again to confirm
GLO	Solo / unsolo the selected track
HOME	PERF RESET: every performance effect off
EDIT	KEYS: the keys play the track's voice
SEQ	The links
PLAY	Play the song from its start
REC · SAVE	Record options · load page

COPY (OCT+) + …

bar key	Copy the shown bar there (on the grid: every track)
track, track	Copy a track
pattern, pattern	Copy a pattern (pattern page)
held + knob	Control All (sound page): the knob on every track; let go, then OCT+ reverts, OCT– keeps

LOCKS

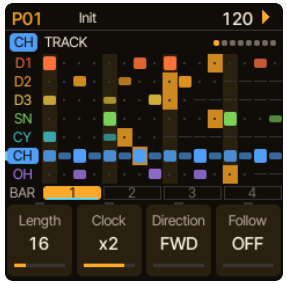
On a sound page, **hold a step key and turn a knob**: that value for that step only. Turning knobs while recording writes locks too.

THE GRID

Bar height: velocity · outline: chance under 100% · notch: flam · tick below: shift · dot top right: lock · dot top left: condition · bar below: ratchet. Knobs: tempo (down to MIDI: follow clock), shuffle, length, volume.

STEP EDIT (SEQ TWICE)

Tap a white key to select a step (again: on / off). SELECT for the pages.



Page	K1	K2	K3	K4
STEP	Velocity	Note	Chance	Condition
TIMING	Flam	Shift	Length	–
LOCKS	Lock 1	value	Lock 2	value
RATCHET	Hits	Shape	Rate	Fade

Length: AUTO, 1/32 … 1/2, TIE (slides into the next note). Shape: even or BOUNCE.

SOUND PAGES

Each page has two rows of four knobs: the button again (or SELECT) gives the other row. MODEL picks LXR, Rings or Plaits (D1–D3) or a MICRO type (any track).

SAVING (SAVE · SHIFT (OCT–) + SAVE LOADS)

KIT	64 slots or the library's 128 · OCT+, name, OCT+
PATTERN	32 · a white key saves to that pattern
SONG	64 · OCT+
MORPH	Saves the mixture you hear; loads as the morph target
SOUND	The library's 128: one track's sound
PROJECT	4 · everything
FACTORY	The 85 factory kits (load only)

Naming: K1 position, K2 letter, K3 case; OCT+ saves, OCT– alone goes back.

PATTERNS AND SONGS (HOME)

Pattern page: white key queues a pattern, bar keys pick the bank. Song page: K1 song, K2 row, K3 the row's pattern (END, LOOP), K4 repeats (x1 … x16).

TRACK PAGES (SEQ ONCE, SELECT)

Page	K1	K2	K3	K4
TRACK	Length	Clock	Direction	Follow
GENERATOR	Generator	its settings		
ACTION	Action	Grids map X / Y, chaos · Euclid's second pulse		
GROOVE	Groove	Amount	Humanize	Follow
BOUNCE	Bounce	Bnc pitch	–	–
MELODY 1	Shape	Spread	Bias	Steps
MELODY 2	Deja vu	Loop	Length	Len var
SCALE	Root	Scale	Follow	–

EACH TRACK ON ITS OWN CLOCK

Clock: /4 · /3 · /2 · 2/3 · 3/4 · **1x** · 4/3 · 3/2 · x2 · x3 · x4. A hat at x2 plays 32nds; 3/4 fits 12 steps in a bar. Add a length of its own (1–64) for polymeters.

FWD · REV	1 2 3 4 1 … · 4 3 2 1 4 …
PING	1 2 3 4 3 2 1 2 … (the ends once)
PING2	1 2 3 4 4 3 2 1 1 2 … (the ends twice)
RAND · WALK	Any step · one step either way

GENERATORS

EUCLID	Length, Hits, Rotate; a second pulse on ACTION (OR, XOR, AND, SUB)
MUTATE	Small changes each time round · Freeze stops it
TURING	Shift register: Length, Lock (127 repeats), Density
CELLULAR	Each time round the next row of rule 0–255
GRIDS	The Grids drum map: Part, Density, Accent

All but EUCLID play a live copy; COMMIT (ACTION page, OCT+) writes it into the pattern.

PERF (ARP): WHITE KEYS ACT WHILE HELD

SHIFT (OCT–) + key latches it. Black keys mute. Bar keys: repeat length 1/4 … 1/32. ALGORITHM picks the throw track.

Key	PERFORM	PERFORM 2
1–7	Roll the track	DRILL: random fast ratchets
8	FILL	HLF: half time
9	RPT: repeat	JIT: jitter
10	FRZ: freeze the delay	CUT: random cuts
11	T.RV: throw to reverb	SHT: decays short
12	T.FX: throw to FX bus	PMP: pump off D1
13	FB+: feedback push	RES: comb resonator
14	STP: tape stop	SPC: reverb pad
15	LP~: low-pass sweep	CRS: crush
16	HP~: high-pass sweep	REV: reverse

PERF PAGES (SELECT)

Page	K1	K2	K3	K4
PERFORM	Tempo	Roll rate	Morph	SRate
MORE	DJ filter	Repeat	Shuffle	Tape time
DENSITY	Density	Ghost vel	Build	Vary
STUTTER	Chance	Size	Length	Vary
2: SETUP 1	Drill	Jitter	Cuts	Pump
2: SETUP 2	Comb	Comb fb	Crush	Reverse

Hold a roll key and turn K2 on PERFORM: a sweep from 1/4 to 1/192.

GLOBAL MOD (ARP THREE TIMES)

LFO 1–3	Dest, Depth, Speed (1/16 … 256 bars), Wave · Mode, Trig, Phase
MACROS	Four knobs, each a parameter between Min and Max (MACRO SETUP)
LINKS	Eight couplings between tracks: When, Target, Effect, Window

SPELLS (ARP TWICE)

A spell changes the sound and the pattern of the tracks you pick, at random but in a direction.

1 Targets	Black keys (bar key 1: all / none)
2 Parts	White keys: what may change
3 Spell	K1 on GRIMOIRE (NONE: plain chance)
4 Cast	COPY (OCT+) · again rolls again
5 Undo / keep	SHIFT (OCT–) alone takes it back · bar key 2 keeps it

Sound (white 1–8)		Pattern (white 9–16)	
1 PIT	pitch	9 HIT	which steps play
2 OSC	osc, FM, models	10 ACC	accents
3 TRN	transient	11 PRB	chance, conditions
4 ENV	envelopes, level	12 TIM	timing, flams
5 FLT	filter	13 RAT	ratchets
6 DRV	drive, shaper, bits	14 NOT	notes
7 FX	sends, pan, insert	15 LEN	length, clock, direction
8 MOD	random, LFO 2, xmod	16 LNK	links, generators

MIXER AND FX

GLO	Volume, pan, FX send, reverb send (2 pages each)
FX send	How much of the voice goes through the FX bus
Reverb send	Feeds the reverb on top of the output
FX	FX bus (delay with TAPE modes, dirt, compressor) → reverb

MIDI

In: USB and the TRS jack · out: USB. Clock out always; tempo down to MIDI follows clock in. CC *n* sets parameter *n* – 1 (as on the LXR), NRPN reaches the rest. Program change 0–31 queues a pattern.